

Gradient descent calculation

$$\Rightarrow y = m_1 x_1 + m_2 x_2 + b$$

$$m_0 = [-1, 2], b_0 = 4$$

$$x(1) = (1, 3), y(1) = 5$$

$$x(2) = (4, 10), y(2) = 6, n = 2, \Delta = 0.01$$

prediction formula: $\hat{y}_i = m_1 x_{i1} + m_2 x_{i2} + b$

Cost function MSE: $J(m, b) = (1/n) \sum (\hat{y}_i - y_i)^2$

Using chain rule: Outer': $\frac{dJ}{d(\hat{y}_i)} = \left(\frac{2}{n}\right) (\hat{y}_i - y_i)$

Inner: $\frac{\partial \hat{y}_i}{\partial m_1} = \frac{x_{i1} \cdot x_{i1}}{x_{i1} \cdot x_{i1} + b} = \frac{x_{i1} \cdot x_{i1}}{x_{i1} \cdot x_{i1} + b}$
 $= 1$

So, $\frac{\partial J}{\partial m_1} = \left(\frac{2}{n}\right) \sum (\hat{y}_i - y_i) \cdot x_{i1}$

$$\frac{\partial J}{\partial m_2} = \left(\frac{2}{n}\right) \sum (\hat{y}_i - y_i) \cdot x_{i2}$$

$$\frac{\partial J}{\partial b} = \left(\frac{2}{n}\right) \sum (\hat{y}_i - y_i)$$

this gives rule: $m_1 \leftarrow \frac{\partial J}{\partial m_1}$, same for m_2 and b

Iteration 1: Starting from $m = [-1, 2]$, $b = 1$

$$\hat{y}^{(1)} = (-1)(1) + (2)(3) + 1 = 6$$

$$\hat{y}^{(2)} = (-1)(4) + (2)(10) + 1 = 17$$

$$e^{(1)} = 4, \quad e^{(2)} = 11$$

$$\text{Cost: } J = (1/2)(e^1 + e^2) = (1/2)(15) = 7.5$$

$$\text{Gradients: } \frac{\partial J}{\partial m_1} = (1)[10(1) + (2)(4)] = 1 + 8 = 9$$

$$\frac{\partial J}{\partial m_2} = (1)[1(3) + (2)(10)] = 3 + 20 = 23$$

$$\text{Rule } \Delta = 0.01, \quad m_1 = -1 - 0.01(9) = -1.09$$

$$m_2 = 2 - 0.01(23) = 1.97$$

$$b = 1 - 0.01(11) = 0.99$$

Iteration 2: $m = [-1.09, 1.97]$, $b = 0.99$

$$\hat{y}^{(1)} = (-1.09)(1) + (1.97)(3) + 0.99 = 2.04$$

$$\hat{y}^{(2)} = (-1.09)(4) + (1.97)(10) + 0.99 = 3.78$$

$$e^{(1)} = 2.96, \quad e^{(2)} = -2.22$$

$$\text{Cost } J = (1/2)(-2.96)^2 + (-2.22)^2 = 6.845$$

Gradients: $\frac{\partial J}{\partial m_1} = -11.84$, $\frac{\partial J}{\partial m_2} = -31.08$
 $\frac{\partial J}{\partial b} = -5.18$

thus updates $m_1 = -1.3316$, $m_2 = 1.1808$, $b = 0.9318$

Iteration 3: $m = [-1.3316, 1.1808]$, $b = 0.9318$

predictions: $\hat{y}^{(1)} = (-1.3316)(1) + (1.1808)(2) = 3.1426$

$\hat{y}^{(2)} = (-1.3316)(4) + (1.1808)(10) = 7.4134$

$e^{(1)} = -1.8574$, $e^{(2)} = 7.4134 - 6 = 1.4134$

Cost $J = \left(\frac{1}{2}\right)[(-1.8574)^2 + (1.4134)^2] = 2.7238$

Gradients: $\frac{\partial J}{\partial m_1} = 3.7962$, $\frac{\partial J}{\partial m_2} = 8.5618$, $\frac{\partial J}{\partial b} = -0.44$

thus updates: $m_1 = -1.3695$, $m_2 = 1.095$, $b = 0.936$

Iteration 4: $m = [-1.3695, 1.095182]$, $b = 0.93624$

$\hat{y}^{(1)} = (-1.3695)(1) + (1.095182)(2) + (0.93624) = 2.752224$

$\hat{y}^{(2)} = (-1.3695)(4) + (1.095182)(10) = 6.409812$

Errs: $e^{(1)} = 2.752224$, $e^{(2)} = 0.409812$

$$\text{Cost } J = \left(\frac{1}{2}\right) [(-2.1477)^2 + (0.4077)^2] = 2.3904$$

$$\text{Gradients: } \frac{\partial J}{\partial m_1} = -0.5882, \quad \frac{\partial J}{\partial m_2} = -2.34, \quad \frac{\partial J}{\partial b} = -1.73$$

This updates: $m_1 = -1.3644$, $m_2 = 1.1196$, $b = 0.953620$

Tread discussion:

Cost J across iterations: $6.1 \rightarrow 6.745 \rightarrow 2.724 \rightarrow 2.390$

The error is dropping sharply then slowing down, this is the classic gradient descent pattern where big early corrections give away to fine tuning as we approach a minimum. ~~m_1 and b are~~ m_1 and b are also converging toward stable values (m_2 settling near -1.37 , m_2 near 1.12 and b near 0.95). Rather than oscillating or diverging, so the updates are moving in a direction that reduces the error.